Did You Get the Level You Asked For?

Bridging LLMs and PCGRL for *Super Mario Bros.* Level Generation

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AIIDE 2026

22nd AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment

Scan to play with the interactive demo →

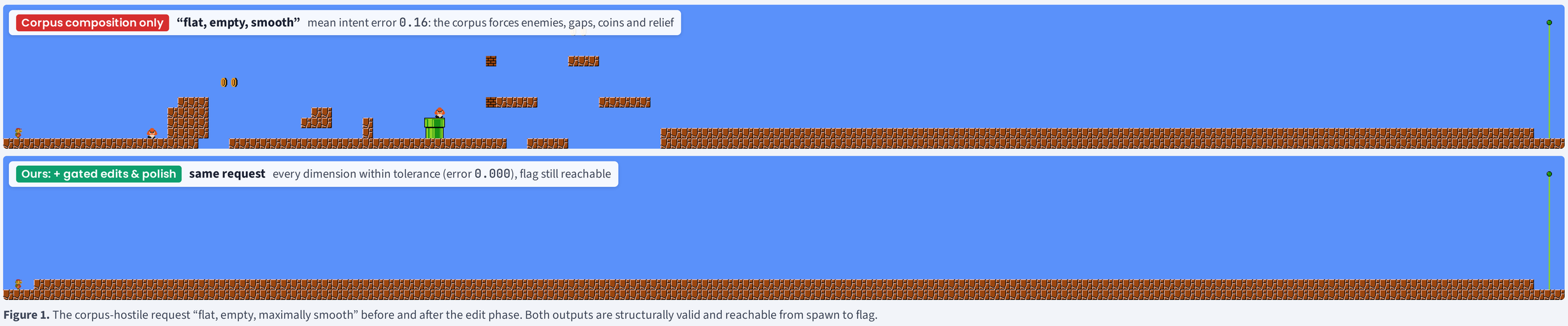


Figure 1. The corpus-hostile request "flat, empty, maximally smooth" before and after the edit phase. Both outputs are structurally valid and reachable from spawn to flag.

1 Motivation

Designers describe levels by **intent** ("a flat, easy stretch", "a hard climb over gaps"), and turning such requests into faithful, playable content is the central control problem of PCG [1,2]. LLMs made natural-language requests easy [3,4], but a corpus-trained generator asked for *zero* enemies still returns a generic level with enemies, gaps and bumps.

0 / 241
middle corpus chunks are flat and empty

190 / 264
chunks contain at least one enemy

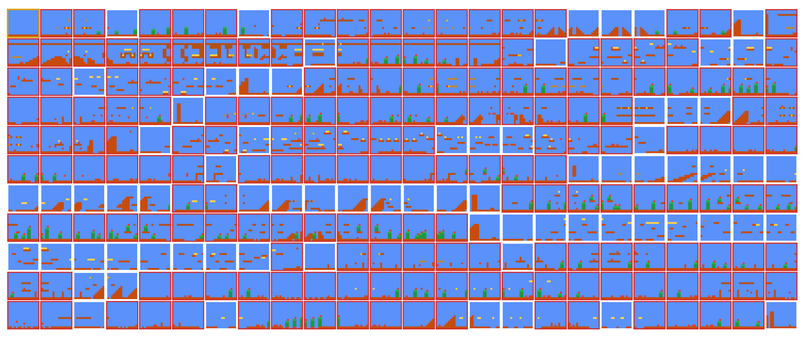


Figure 2. All 264 corpus chunks. **Red**: contains an enemy (190). **Gold**: the only flat, empty chunk, a *start* chunk.

The issue is not obedience but **material**: if every corpus pattern contains an enemy, no recombination contains none. Generators are **material-bound**.

Goal. Make intent satisfaction explicit, measurable and reachable, *including the 0 and 1 boundaries*, while keeping corpus style and verified playability.

2 Contributions

- A two-phase **compose-and-edit MDP**: one policy assembles corpus chunks, then reshapes the level with gated macro-edits.
- A deterministic **polish** stage that inverts the metrics and reports per-dimension residuals.
- An input-only **LLM bridge** with record/replay, so translation and generation error are measured separately.
- Evaluation against search baselines, boundary sweeps, ablations and an external **A*** judge.

3 An eight-number design contract

Intent is a vector $\mathbf{t} \in [0,1]^8$ measured on a 152 × 14 tile grid.

Table 1. The eight intent dimensions and their normalizers (p95 cap: corpus 95th-percentile density; band: achievable-band rescaling).

enemies	goombas per area	p95 cap
coins	coins per area	p95 cap
blocks	? blocks, bricks, used blocks	p95 cap
pipes	pipe tiles per area	p95 cap
gaps	width-weighted, jump-relative gap count	1 gap / 24 col.
verticality	amplitude of the reachable path envelope	band
smoothness	1 − mean envelope step / jump height	band
platforms	share of the path above ground	band

Achievable-band re-normalization. Raw structural metrics occupy a narrow band (e.g. smoothness ∈ [0.64, 0.93]). We rescale them to what legal compositions can reach:

Eq. 2 Achievable-band rescaling of a structural metric

$$\hat{m} = \mathrm{clip}\left(\frac{m - q_{05}}{q_{95} - q_{05}},\ 0,\ 1\right)$$

so 0 means *as little as the medium permits*. Platform-usage translation error fell from 0.137 to **0.053**.

4 Method

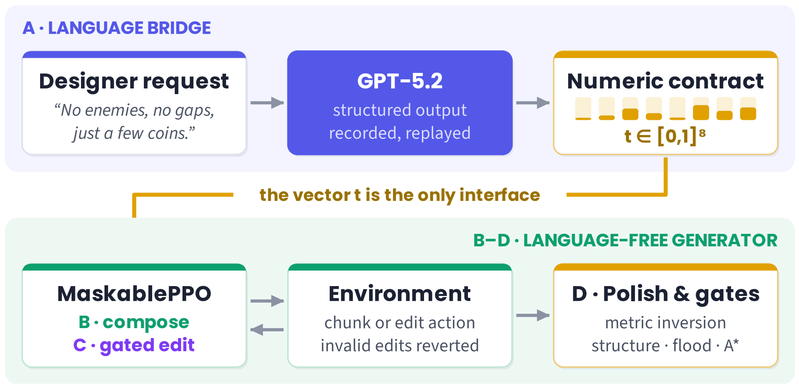


Figure 3. Language defines **t**; a language-free policy builds and edits the level; polish and three gates run before delivery.

A Language bridge. GPT-5.2 maps the request to **t** through a structured-output prompt with a glossary and defaults. Translations are content-addressed and replayed exactly. **It never places a tile.**

B Compose. A MaskablePPO policy [7] appends ~10 of 264 annotated 16-column chunks from 15 VGLC levels [6]. Hard masks enforce seam physics, family order and width.

C Gated edit. The *same* policy then picks macro-edits: 16 types × 8 segments (remove or place enemies, coins, blocks, pipes; fill or dig gaps; flatten or raise terrain; platforms). Each edit is checked by validators and a spawn-to-flag flood; **failed edits are reverted** and still cost one of 24 decisions.

D Polish & gates. Deterministic metric inversion (tolerance 0.05) reuses the edit planners. Release requires structure, reachability and an **A* agent** [8] win with ≥ 0.95 completion.

Eq. 1 Terminal reward · one signal for both phases (γ = 1)

$$R = 4\ \underbrace{align}_{\text{match the target}} + 2\ \underbrace{style}_{\text{corpus-like}} + 3\ \underbrace{play}_{\text{reachable}} - \underbrace{dup}_{\text{no repeats}}$$

$$align = 0.6\big(1 - \mathrm{mean}\,|\delta|\big) + 0.4\big(1 - \max|\delta|\big) \qquad style = \exp\big(-\mathrm{TPKL}(\mathrm{level}\,\|\,\mathrm{corpus})\big)$$

$$play = \begin{cases} 1 & \text{flag reached} \\ 0.5\,c/c_{\mathrm{flag}} & \text{otherwise} \end{cases} \qquad dup = \tfrac{1}{n-1}\textstyle\sum_{k\geq 2} 1\left[x_k \in \{x_{k-3..k-1}\}\right]$$

δ = measured − target · c = rightmost column the flood reaches · x_k = k-th of n chunks

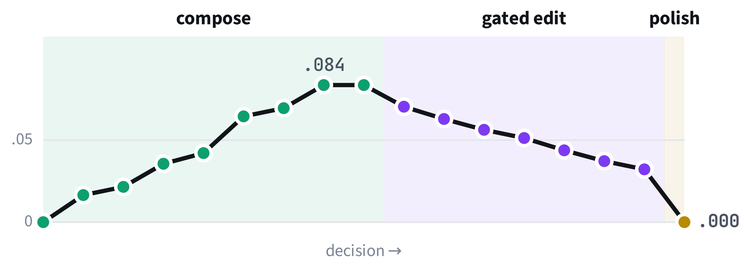


Figure 4. One real final-policy episode for "flat, empty, smooth": corpus chunks raise the intent error; gated edits and polish bring it to zero.

5 From language to level

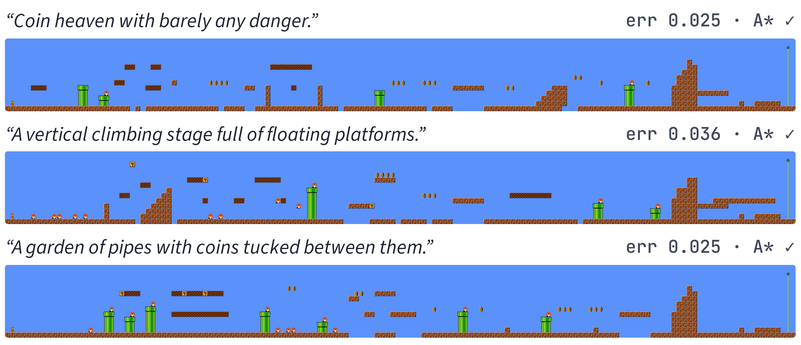


Figure 5. Recorded GPT-5.2 translations, generated end-to-end by the final policy and completed by the A* agent.

Edits go where they survive. A reverted edit still spends one of the 24 decisions, so the policy edits only when the target demands it (~15 decisions per episode), and about a third of accepted edits land in the sparse final segment.

6 Results

RQ1 Does learning beat search?

Yes. Over the 36-target suite, PPO beats width-8 beam search in the same chunk space on all three seeds by **+0.97 to +1.03** reward at best-of-5 ($p = 10^{-4}$), halving intent error, at **64 ms vs 0.47 s** per level.

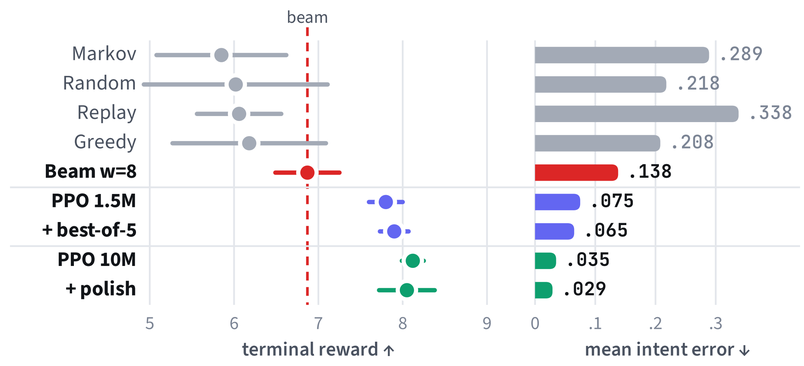


Figure 6. Terminal reward (mean ± std) and mean intent error on the 36-target suite; dashed line: beam search.

RQ2 Is control accurate at the boundaries?

The raw 10M-step policy passes **all 8 sweeps** from 0 to 1, including exact-zero enemies and gaps. Polish makes counts near-exact but trades smoothness away: a residual the system **reports**.

Table 2. Final-policy sweep MAE per dimension (targets 0 to 1, 10 levels each); pass ≤ 0.10. * Reported residual (metric tension).

dimension	raw	+polish	dimension	raw	+polish
enemies	.048	.017	gaps	.029	.019
coins	.036	.015	verticality	.073	**.103***
blocks	.038	.017	smoothness	.057	**.141***
pipes	.055	.022	platforms	.029	.046

RQ3 What does editing add?

On ordinary targets, little. On **corpus-hostile** targets, compose-only error rises 48% while the two-phase policy degrades far less ($p = 0.0095$); polish reaches **0.004** on flat-empty.

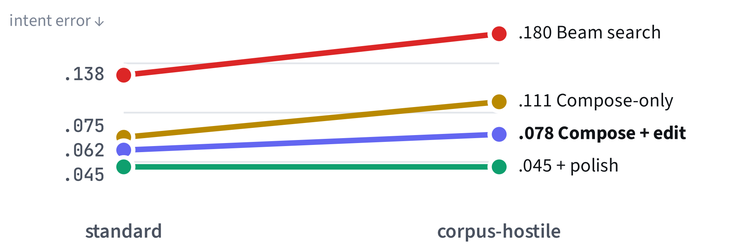


Figure 7. Mean intent error on standard vs corpus-hostile targets (medium tier, best-of-5).

RQ4 How much error does language add?

Translation MAE **0.053** [0.042, 0.067]; end-to-end response is monotonic (ρ = 0.97–0.99). Error grows 0.035 → 0.081: **translation is the bottleneck**.



Figure 8. Measured density vs the intensity word in the request (final model; ρ: Spearman correlation).

Judge Is geometric reachability enough?

No. Calibrating the training-time flood against the A* agent on 120 levels:

0.758
best flood–agent agreement (jump 4 × reach 5)

34%
of "reachable" levels the agent still fails (enemies, timing)

Hence the two-tier protocol: flood for training and sweeps, **agent completion** for release.

7 Limitations

- One game, 15 overworld levels; new themes need new operators and validators.
- Metrics describe tile structure, not difficulty or fun; no human study.
- Flood ≠ playability (agreement 0.758), hence the A* gate; A* is not a human.
- The 10M-step model has one seed; one LLM evaluated.

8 Conclusions

- The request becomes a **numeric contract**: interpretation and fulfillment are tested separately.
- Editing is unnecessary for ordinary targets but **decisive** once a request leaves corpus support.
- Every miss is **attributable**: translation, generator, or conflicting targets.

9 Future work

- Remove the fixed vocabulary: the LLM proposes **new level properties**, compiled into checkable metrics.
- Let language **edit any aspect** of the level, under the same gates and residual reports.
- Goal of the master's thesis: open-ended requests that stay **verifiable** and attributable.

References. [1] Khalifa et al. PCGRL: Procedural Content Generation via RL. AIIDE 2020. [2] Earle et al. Learning Controllable Content Generators. CoG 2021. [3] Sudhakaran et al. MarioGPT: Open-Ended Text2Level Generation through LLMs. NeurIPS 2023. [4] Todd et al. Level Generation Through Large Language Models. FDG 2023. [5] Baek et al. IPCGRL: Language-Instructed RL for Procedural Level Generation. CoG 2025. [6] Summerville et al. The VGLC: The Video Game Level Corpus. 2016. [7] Huang & Ontañón. A Closer Look at Invalid Action Masking in Policy Gradient Algorithms. FLAIRS 2022. [8] Karakovskiy & Togelius. The Mario AI Benchmark and Competitions. IEEE TCIAIG 2012.

Acknowledgments. Partially funded by CAPES, CNPq and Fapemig.

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Code, trained models & data github.com/brunolemosdl/intent-pcgrl